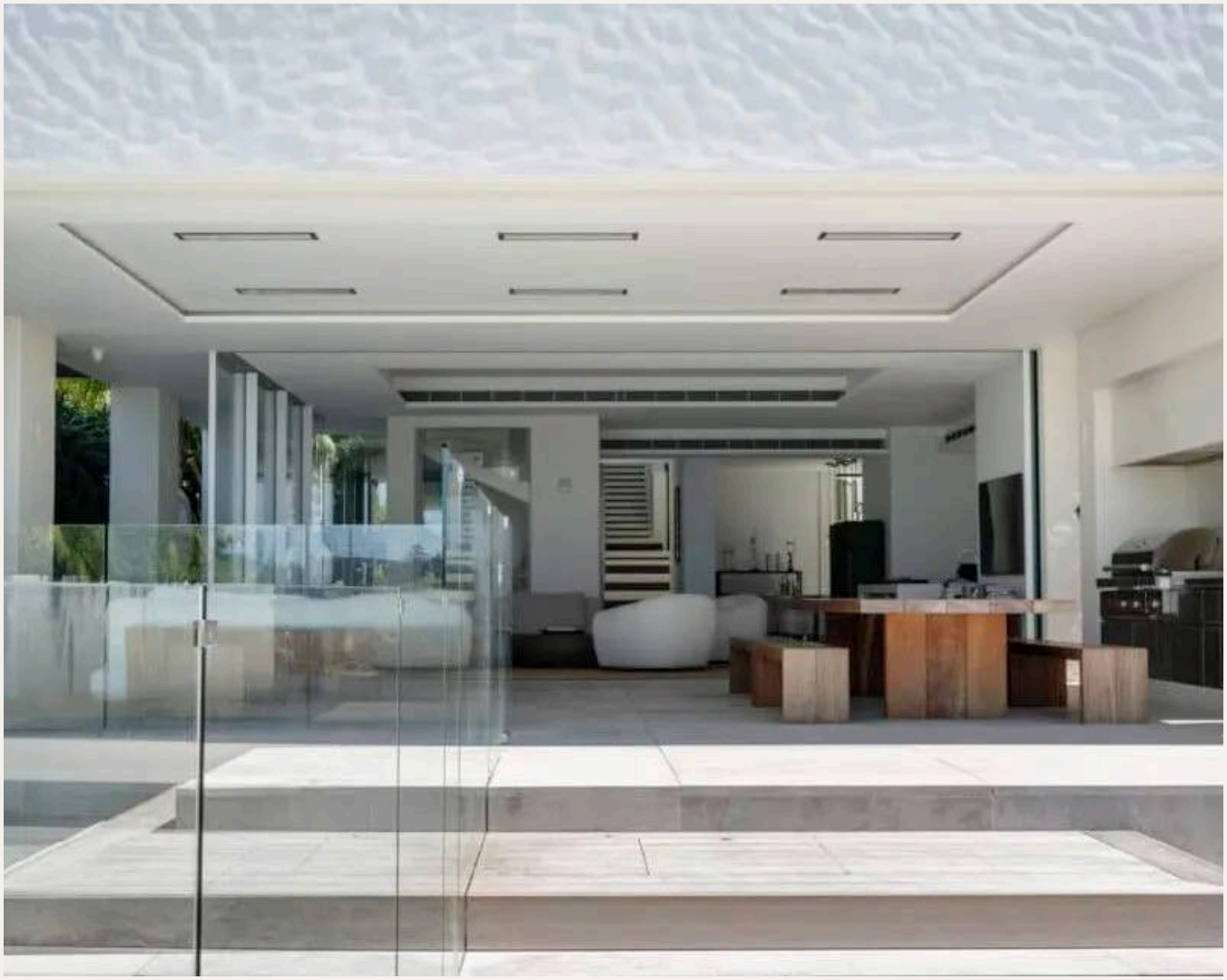
BROMIC
HEATING

ARCHITECT / DESIGNER · BUILDER / INSTALLER

The outdoor heating planning guide: professional edition

Specifier's guide to outdoor radiant heat: wavelength, the current range by model, amps at 208/240 V, controller channels, clearances and code.



EDITION
October 2026

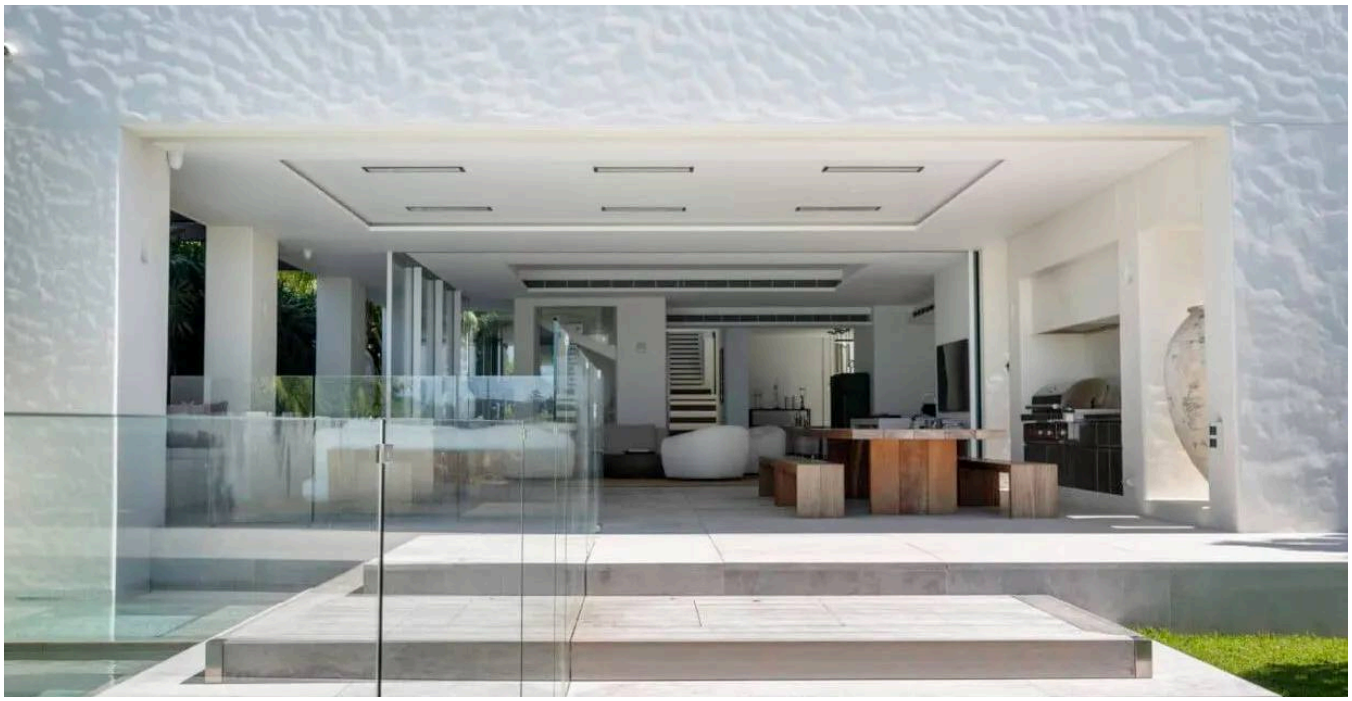
BROMIC HEATING
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The outdoor heating planning guide

Professional edition, for architects, designers and builders. Designed from the start, not added later.

Short wave heaters deliver about 90% of their output as radiant heat and warm instantly; medium wave about 60% with a 30-second warm-up; long wave about 40% with a 5-minute warm-up and the most sensitivity to wind[3]. That choice, made with the fuel, the mounting and the load, belongs in schematic design. This edition runs the sequence in the order a drawing set needs it, with the data per model.

Fewer than one in four homeowners prioritize heating at planning[1]. Late heat arrives as surface conduit, a mismatched ceiling and a substitution at bid. Heat in the set from schematic design is a line item, a circuit and a recess, and it stays.



Minimal glass pavilion with ceiling-recessed heaters

The sequence



STEP	OUTPUT FOR THE DRAWINGS
1 Is it outdoor?	Configuration, open perimeter in %
2 Measure	Heated zones over seating, ft² per zone, mounting surfaces and heights
3 Choose fuel	Electric or gas per zone; wavelength
4 Style and mounting	Model, finish, mount type, accessories
5 Size it	Count per zone (A03 calculator)
6 Power or gas	Amps per heater, circuits, gas load and supply
7 Controls	Controller and channels per zone, wall switch, app or BMS
8 Clearances and code	Heights, clearances, certification, permit path
9 Design service	Layout and coverage drawing from Bromic (A07)

1. Is it outdoor?



An area is outdoor for heating design when at least 30% of its horizontal perimeter is permanently open. A partial overhead cover holds a heat cloud that adds a convective contribution; size on radiant coverage regardless[3].

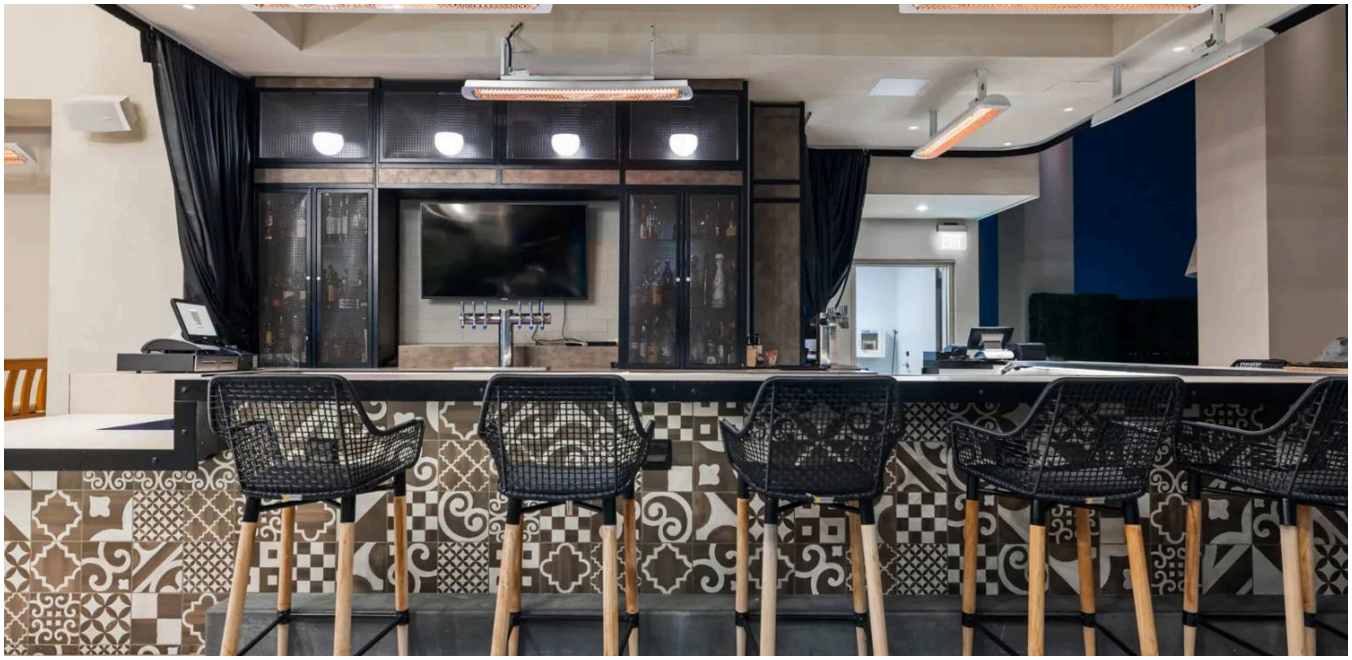
CONFIGURATION	QUALIFIES	DESIGN NOTE
Walls all round, one permanent ground-level opening, no roof	Yes, if the opening is 30% or more of the perimeter	No heat cloud; size for exposure
Overhead cover, two open sides	Yes	Heat cloud under cover
Overhead cover, three walls, 30% or more open	Yes	Strong heat cloud; check clearances
Retractable glazing or screens that close below 30%	Check	Treat as enclosed; confirm with the AHJ and Bromic

Rules per the 2021 guide[3].

DIAGRAM: Plan and section of the four configurations, perimeter opening shaded and dimensioned as a percentage.

- ☐ Open perimeter _____% (permanent openings only)[3]
- ☐ Future enclosure noted on the drawings

2. Measure



- [ ] Heated zones drawn over seating and bar rails, not circulation; ft² per zone
- [ ] Mounting surfaces: soffit, beam, wall, post; finished heights AFF
- [ ] Ceiling build-up and cavity depth where recessing
- [ ] Exposure: prevailing wind, water, altitude

3. Choose fuel

Outdoors, convected heat rises and disperses; forced air cannot hold an open space at temperature and outdoor heat cannot be thermostatically controlled[3].

	SHORT WAVE	MEDIUM WAVE	LONG WAVE
Radiant / convective split	about 90 / 10	about 60 / 40	about 40 / 60
Warm-up	Instant	about 30 s	about 5 min
Wind	Lowest sensitivity	Limited	Highest
Use	Exposed sites, higher mounting	Covered areas	Sheltered, low-exposure spaces

Source: Bromic Planning Guide 2021[3]. Directional heaters push most output toward the occupied zone, which puts the energy where people sit[3].

	ELECTRIC	GAS
Space	Covered and semi-covered; soffits, walls, poles	Larger, more open areas; freestanding
Supply	Dedicated branch circuits; 208, 240 or 277 V by model	Natural gas or propane, plus 110 V 2 A for built-in gas[1011]
Integration	Recessed, flush, pendant	Surface; deflectors reduce top clearance
Control	On/off or 1% dimming, zones, schedules, app, BMS	On/off, zones, schedules, app, BMS
Trade	Licensed electrician	Licensed gas fitter and electrician

Jurisdictions restricting new gas: see the Electrification compliance guide (A12).

Urban roof terrace, timber soffit with recessed heater

4. Style and mounting: the current range

MODEL	RATING	COVERAGE	MOUNT	IP / CERTIFICATION	WARRANTY
Tungsten Smart-Heat Electric	2000 / 3000 / 4000 / 6000 W; 220–240 V, 208 V, 277 V on 4000/6000	65 / 100 / 130 / 160 ft²	Recessed, ceiling, wall	IPX4; ETL/cETL 4005951	2 yr heater, 7 yr elements[4]
Platinum Smart-Heat Electric	2300 / 3400 / 4500 W; 220–240 V and 208 V	65 / 108 / 143 ft²	Wall, pole (8 in included), recessed	ETL/cETL 5001799	3 yr[5]

MODEL	RATING	COVERAGE	MOUNT	IP / CERTIFICATION	WARRANTY
Platinum Smart-Heat Electric Marine	2300 / 3400 / 4500 W; 220–240 V or 208 V	65 / 108 / 143 ft²	Wall, pole, recessed	IPX5; Lloyd's Register marine approval	5 yr land, 3 yr vessel[6]
Eclipse Smart-Heat Electric	2900 W; 0/33/66/100%	121 ft²	24 in twin pole; 8/24/48 in straight, curved, wall-mount poles	SGS	3 yr[7]
Eclipse Smart-Heat Freestanding	2900 W; 220–240 V	121 ft²	Freestanding	SGS	3 yr[8]
Cobalt Smart-Heat Electric	4000 / 6000 W	130 / 160 ft²	Wall, ceiling	IPX4; ETL/cETL 4005951	1 yr parts, 7 yr elements[9]
Platinum Smart-Heat Gas	23,600 / 39,800 Btu/h; NG or LPG	160 / 215 ft²	Wall or ceiling; poles available	Intertek 4009079	2 yr excl. hoses, igniters[10]
Tungsten Smart-Heat Gas (built-in)	Two models; output not published here	160 / 215 ft²	Wall or ceiling; pivot arm ±25°		2 yr excl. hoses, igniters[11]
Tungsten Smart-Heat Freestanding Gas	38,000 Btu/h; LPG 20 lb cylinder or NG quick-connect	215 ft²	Freestanding	Intertek 4009079 (LPG)	1 yr (LPG)[12]
Tungsten Electric Coastal (coming soon)	; 316 stainless	64–144 ft²	Marine-grade low-clearance recessed, ceiling, wall, single-post		[2]
Tungsten Electric Select (coming soon)		64–144 ft²	Low-clearance recessed, ceiling, wall, single-post		[2]

SKUs, Platinum Electric: 2300 W BH0320003 black / BH0320007 white; 3400 W BH0320005 / BH0320008; 4500 W BH3622000 / BH3622001[5]. Platinum Gas: BH0110001-1 NG / -002-1 LPG (23,600 Btu/h); BH0110003-1 NG / BH0110004-1 LPG (39,800 Btu/h)[10]. Tungsten Gas built-in: BH0210001-1 NG / BH0210002-1 LPG (160 ft²); BH0210003-1 NG / BH0210004-1 LPG (215 ft²)[11].

Product-page values read 29 October 2026; Lookbook 2026 governs where they differ[2]. Open conflicts: . The Tungsten built-in gas output is withheld until confirmed.

ACCESSORY	FUEL	PURPOSE
Recess kit	Electric (Platinum, Tungsten)	Flush-mounts into the ceiling cavity

ACCESSORY	FUEL	PURPOSE
Suspension pole	Electric (Platinum, Tungsten)	Drops the heater to an effective height
Heat deflector	Gas (built-in)	Dual-layer stainless; reduces clearance above. Recommended when space above is under 1,100 mm (Platinum) or 42 in (Tungsten), or the unit is exposed[10][11]
Ceiling pole	Gas (built-in)	Two heaters back to back under flat or pitched ceilings[3]

- [ ] Model, wattage or Btu/h, voltage, finish and SKU per tag
- [ ] Coming-soon models carry an approved current alternate
- [ ] Recess kits, poles, deflectors scheduled

Black Tungsten Select grille glowing, macro detail

5. Size it

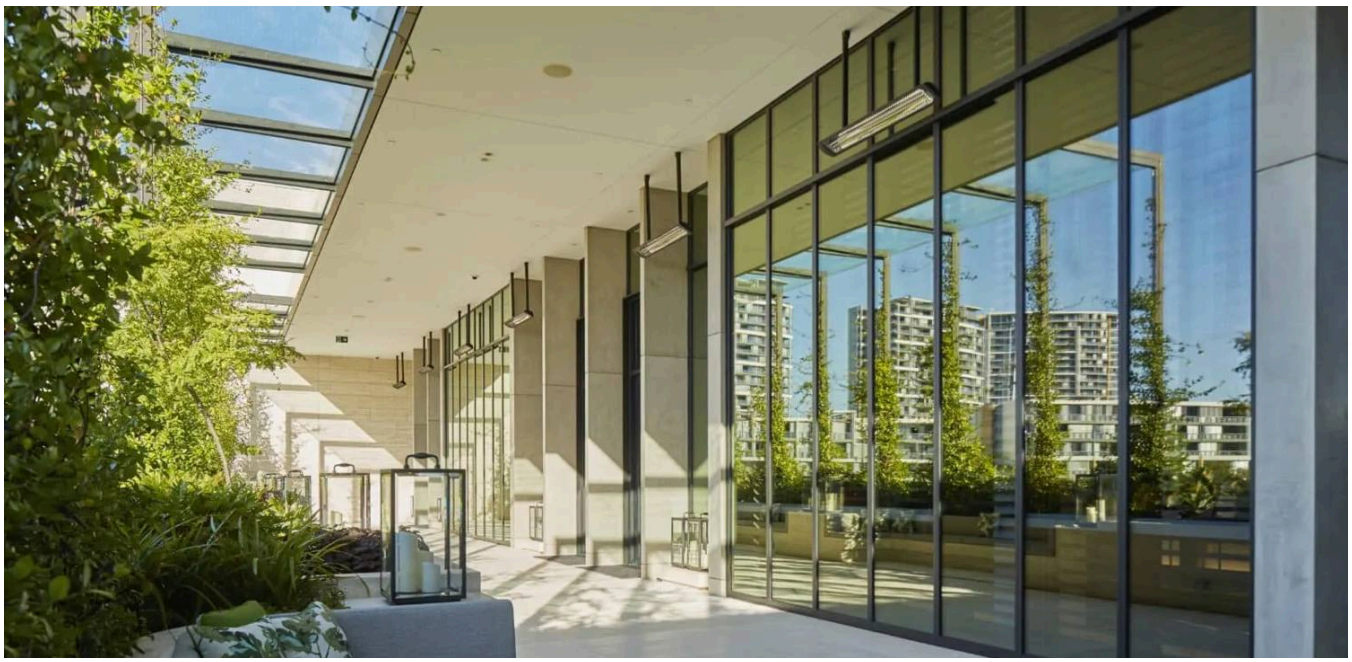


Heaters per zone = zone area ÷ effective coverage per heater, rounded up. The A03 calculator applies its stated adjustments to the nominal coverage above: open space 0.85, partially covered 1.0, covered 1.1; mild 1.0, cool 0.85, cold 0.7; mounting to 10 ft 1.0, to 12 ft 0.9, above 12 ft 0.8 (built-in heaters). These are working assumptions awaiting Bromic sign-off, not published ratings. Freestanding heaters (Tungsten Freestanding Gas, Eclipse Freestanding) also serve no more than four seats each, because seated guests block their radiant heat: count max(area ÷ effective coverage, seats ÷ 4), allowing 20 ft² a seat where the seat count is unknown.

Example. A 16 × 24 ft covered dining terrace, cool climate, 10 ft soffit: 384 ft². Tungsten Electric 6000 W: 160 × 1.1 × 0.85 × 1.0 = 150 ft² effective; 384 ÷ 150 = 2.6, so **three heaters**, each on its own Affinity Dimmer channel (a 5-channel controller), a wall switch and the app.

- [ ] Count per zone with factors stated
- [ ] Heat blooms laid over seating

6. Power or gas



Current: I = W ÷ V. Fixed electric space-heating loads are treated as continuous, so branch-circuit conductors and overcurrent devices are sized at not less than 125% of the load (NEC 424 with 210.19(A)(1) and 210.20(A))[2421]. Feeders and services follow Article 220.

HEATER	A AT 208 V	× 1.25	A AT 240 V	× 1.25	A AT 277 V
Platinum 2300 W	11.1	13.9	9.6	12.0	—
Platinum 3400 W	16.3	20.4	14.2	17.8	—
Platinum 4500 W	21.6	27.0	18.8	23.5	—
Eclipse 2900 W	13.9	17.4	12.1	15.1	—
Eclipse Freestanding 2900 W	—	—	12.1	15.1	—
Cobalt 4000 W	18.75	23.4	16.6	20.8	—
Cobalt 6000 W	28.4	35.5	25	31.3	—
Tungsten 2000 W (calc.)	9.6	12.0	8.3	10.4	—
Tungsten 3000 W (calc.)	14.4	18.0	12.5	15.6	—
Tungsten 4000 W (calc.)	19.2	24.0	16.7	20.8	14.4
Tungsten 6000 W (calc.)	28.8	36.1	25.0	31.3	21.7

Published amps from the product pages[5789]; Tungsten rows are W ÷ V because the page lists none. Cobalt's published 208 V figures sit below W ÷ 208 (19.2 and 28.8 A).

Gas. Size piping for the total connected Btu/h, length and pressure to ANSI Z223.1/NFPA 54[22]. Built-in gas heaters also need 110 V, 2 A[1011]. The Tungsten Freestanding Gas takes a 20 lb LPG cylinder or a 10 ft natural-gas quick-connect[12]. Do not carry a Tungsten built-in Btu/h into drawings until Bromic confirms it.

Rough-in detail (boxes, blocking, control runs): Pre-wire guide (A16).

- ☐ Amps per heater at the specified voltage; conductors and OCPD at 125% or more[24]
- ☐ Panel capacity; junction boxes dimensioned
- ☐ Gas load, pipe size, shut-offs; 110 V at each built-in gas heater

7. Controls (required)

Every heater schedule carries a controls line. Specify the full bundle: controller and channels, wall switch, app or BMS.

ITEM	CHANNELS	CAPACITY PER CHANNEL	SPECIFY WITH
Affinity Smart-Heat On/Off	1, 2, 5	6 kW at 208–277 V (1 kW at 100–127 V); up to 10 gas heaters	Electric and built-in gas[13]
Affinity Smart-Heat Dimmer	1, 2, 5	6 kW at 208–277 V; 1% steps	Tungsten and Platinum Electric[14]
Affinity Smart-Heat Eclipse	1, 5	One Eclipse heater	Eclipse[15]
Manual wall switch	Single or dual	20 A, 120–277 V; indoor or outdoor kit	All[16]
Expansion board	—	Ethernet and 0–10 V for BMS or home automation; built into the 5-channel; BMS mode disables the app	Integrated projects[17]

Heaters per 6 kW channel: Tungsten 2000 W ×3, 3000 W ×2, 4000/6000 W ×1; Platinum 2300 W ×2, 3400/4500 W ×1; Cobalt ×1. Eclipse: one per Eclipse channel. Gas: up to 10 per On/Off channel[13]. The circuit rating can set a lower limit.

Controllers: IP54, operating range −4 to 86 °F; 2.4 GHz Wi-Fi, Bluetooth about 30 ft, optional Ethernet; UL 60730-1, CAN/CSA E60730-1[13]. Wi-Fi and app features require compatible Affinity Smart-Heat Controls, sold separately. Alexa and Google Home need no board; Control4, Crestron and Lutron compatibility varies and may need extra hardware or configuration[217].

- ☐ Controller, channel count and heater-to-channel schedule

- [ ] Controller location: accessible, protected, within Wi-Fi range
- [ ] Wall-switch box at the threshold
- [ ] App or BMS; integrator named

Eclipse with dimmable rim lighting glowing

8. Clearances and code

MOUNT	MINIMUM AFF
Tungsten, Platinum, Platinum Marine Electric	Wall 72 in; ceiling 96 in[4][5][6]
Cobalt Electric	Wall 6 ft; ceiling 8 ft[9]
Eclipse Electric	96 in[7]
Eclipse Freestanding	79 in clearance[8]
Platinum Gas, Tungsten Gas built-in	96 in; Tungsten optimum 8 ft[10][11]

SCOPE	US STANDARD	GOVERNS
Electric	ANSI/NFPA 70[21]	Circuits, overcurrent protection, disconnects, wiring methods
Gas	ANSI Z223.1/NFPA 54[22]	Where heaters may go, supply, ventilation, clearance to combustibles
Certification	Intertek, UL or CSA mark	Required on gas and electrical appliances, including portable heaters[3]

Clearances to ceiling, sides and combustibles come from each model's installation manual and listing; do not interpolate between models.

Indicative only. The adopted code edition and local amendments govern. Confirm with the local building authority, and have heaters installed by a licensed electrician or gas fitter to the manufacturer's instructions.

- ☐ Heights AFF meet the minimums
- ☐ Per-model clearances on the drawings
- ☐ Certification mark per model; permits and AHJ path identified

White heater between pergola louvres

9. Design service

Send plans, sections or photos to the Bromic design service for heater selection, placement and coverage drawings, at no charge[2]. Running cost per heater: kW × $/kWh (electric); Btu/h ÷ 100,000 × $/therm (natural gas); Btu/h ÷ 91,500[23] × $/gal (propane). At the A20 model's rates (13.54¢/kWh, EIA commercial, May 2026[18]; $0.87/therm, EIA commercial, Jan–May 2026[19]; $2.674/gal, EIA residential propane, 30 Mar 2026[20]) a Tungsten 4000 W costs $0.54 an hour and a Platinum Gas 39,800 Btu/h $0.35 on natural gas or $1.16 on propane.

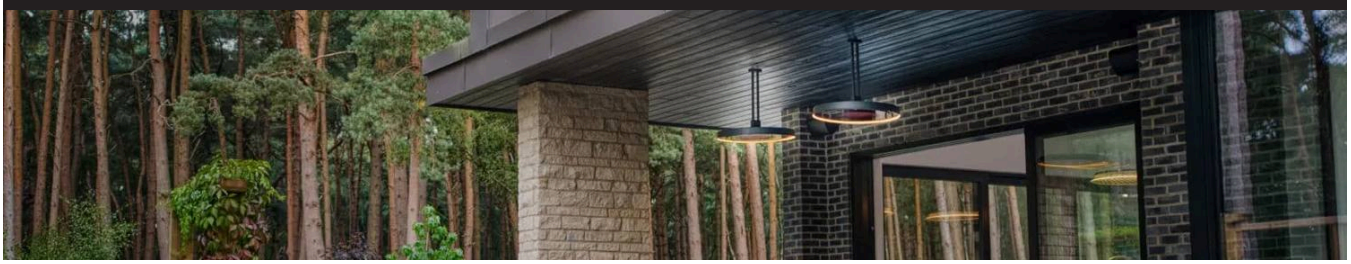
- ☐ Drawings sent to the design service
- ☐ Coverage drawing returned and coordinated

Tungsten Coastal (coming soon) recessed flush into a white ceiling

Download the spec kit for CSI sections, details and coverage tables.

NEXT STEP

Download the spec kit



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Confirm every installation with your local building authority and a licensed electrician or gas fitter. Wi-Fi and app features need compatible Affinity Smart-Heat controls, sold separately. © 2026 Bromic Heating · bromic.com